Subspace Clustering Constrained Sparse NMF for Hyperspectral Unmixing

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***Abstract*— As one of the most important information of hyperspectral images (HSI), spatial information is usually simulated with the similarity among pixels to enhance the unmixing performance of nonnegative matrix factorization (NMF). Nevertheless, the similarity is generally calculated based on the Euclidean distance between pairwise pixels, which is sensitive to noise and fails in capturing subspace information of hyperspectral data. In addition, it is independent of the NMF framework. In this article, we propose a novel unmixing method called subspace clustering constrained sparse NMF (SC-NMF) for hyperspectral unmixing to more accurately extract endmembers and correspond abundances. First, the nonnegative subspace clustering is embedded into the NMF framework to learn a similar graph, which takes full advantage of the characteristics of the reconstructed data itself to extract the spatial correlation of pixels for unmixing. It is noteworthy that the similar graph and NMF will be simultaneously updated. Second, to mitigate the influence of noise in HSI, only the k largest values are retained in each self-expression vector. Finally, we use the idea of subspace clustering to extract endmembers by linearly combining of all pixels in spectral subspace, aiming at giving a reasonable physical significance to the endmembers. We evaluate the proposed SC-NMF on both synthetic and real hyperspectral data, and experimental results demonstrate that the proposed method is effective and superior by comparing with the state-of-the-art methods.**

***Index Terms*— Hyperspectral unmixing, self-expression, spatial structure, subspace clustering.**

I. Introduction

With the development of remote sensing technology, the spectral resolution of hyperspectral sensors is constantly increasing [1]–[5]. Unfortunately, they have low spatial resolution, which causes a mixture of pixels in images. This phenomenon widely exists in hyperspectral images (HSI), which seriously hinders the application of HSI [6]–[8]. Therefore, to make full use of hyperspectral data, unmixing is proposed to identify the pure spectral features (*endmembers*) and their proportions (*abundances*) in each pixel [9], [10].

A variety of unmixing methods based on a *linear mixing model* (LMM) have been developed in the past decades [11]. LMM assumes that each pixel in the image is linearly combined by its mutually independent endmembers [12]. Under LMM, unmixing techniques can generally be divided into three categories: geometry-based, sparse-regression-based, and statistics-based [6], [13], [14].

The geometrical algorithms are based on convex geometry theory, which assumes that hyperspectral data exist in a convex simplex, and pure materials are located exactly at the vertices of the simplex. Among these algorithms, some methods assume that there are always pure pixels in the HSI. The classic methods are: *pixel purity index* (PPI) [15], *vertex component analysis* (VCA) [16], and N-FINDR [17]. In this case, if the vertices of the simplex are found, the endmembers are extracted. Unfortunately, considering the practical factors, especially in highly mixed data, pure pixels in hyperspectral data do not generally exist [18], [19]. Thus, some methods based on the image itself are proposed, such as *minimum volume simplex analysis* (MVSA) [20], *minimum volume enclosing simplex* (MVES) [21], and *iterative constrained endmembers* (ICE) [22]. It is worth noting that most of the geometry-based methods can only be used to extract endmembers [14]. Therefore, some abundance estimation algorithms are also necessary, such as the fully constrained least squares (FCLS) [23] and simplex projection unmixing [24].

Sparse regression methods, by assuming that pixels can be expressed in the form of linear combinations of some known pure signatures in advance, have shown very strong effectiveness in the hyperspectral unmixing [25]–[28]. They treat unmixing as the problem with supervised information, which greatly reduces the difficulty of endmembers extraction. However, this type of method relies on the correct prior knowledge. Since it must know a dictionary that contains all the endmembers in the image before unmixing, to a certain degree, it will bring poor algorithm performance. For example, there may be a bias between the dictionary and the real endmember spectrum, or the dictionary does not completely contain pure substances of the image [29].

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Under the statistical framework, hyperspectral unmixing can be formulated as the class of blind source separation problems [30]. The popular algorithms include *independent component analysis* (ICA) [31], *nonnegative matrix factorization* (NMF) [32], and the Bayesian approaches [33]. It is particularly noteworthy that NMF is widely used in HSI unmixing with outstanding advantages. NMF decomposes data by seeking two nonnegative matrices [34]. One is used to learn parts-based representation of the raw data, and the other is the coefficient of the base vector. According to these characteristics, it can be seen that NMF is suitable for HSI unmixing. Nevertheless, NMF is usually a local optimal solution since the objective function is nonconvex [14]. To improve the unmixing performance, various matrix factorization methods with additional constraints have been proposed [35]. Sparsity and smoothness properties of hyperspectral data are added as important constraints to the NMF framework in [36]. For instance, *minimum volume-constrained NMF* (MVCNMF) takes the simplex volume into consideration and introduces a weighted sparse regularizer [37]. Among all sparse constraints, L_0-regularization is one of the most commonly used methods, but it is easy to generate an NP-hard problem [38]. Thus, there are some L_p regularizers designed to produce sparse results. Guo *et al.* [39] and Bioucas-Dias and Figueiredo [40] proposed the L_1 regularizer, and $L_{1/2}$ also has also been proven to be very effective for sparse constraints, especially in [41] and [42]. $L_{1/2}-$NMF extension algorithms that have been introduced in [43]–[45]. In addition to these, Guan *et al.* [46], [47] also propose a series of NMF framework during the research process. Gao *et al.* [48] show a method of adaptive fine-tuning of NMF using the variational approach to extract more accurate basis features. *NMF with regularizations* (RNMF) considers the local manifold smoothness and a rank constraint to find a better representation of data [49]. However, most of these techniques are only constrained by the result matrix after unmixing, largely ignoring the specifics inherent in hyperspectral data, especially spatial structure information.

Recently, many researchers have turned their eyes to integrate the spatial information of hyperspectral data into NMF to improve the efficiency of unmixing. For instance, Bioucas-Dias *et al.* [6] propose a sparse NMF method with manifold constraints to exploit the latent structure of the data. *Structure-constrained spare NMF* (CSNMF) is introduced to mine the structure information of hyperspectral data [50]. In [51], *geometric NMF* (GNMF) is proposed, which defines two geometric distances to formulate spatial–spectral manifold regularizer and obtains a more satisfactory result. Zhang *et al.* [52] greatly improve the abundance estimation results, by using both spectral and spatial weighting factors to develops a spectral–spatial weighted sparse unmixing (S^2WSU) method. Nevertheless, the existing methods are not sufficient to fully exploit spatial features among hyperspectral data. Because they usually use the similarity between pairwise pixels to determine spatial information, and the similarity is often expressed based on the Euclidean distance. The Euclidean distance is very sensitive to noise and local neighborhood size in the image. An example illustration of drawbacks of the similarity based on pairwise distance and our idea is shown in Fig. 1. In addition, the similarity is independent of the NMF framework, and its correctness will have a greater impact on the unmixing performance of NMF.

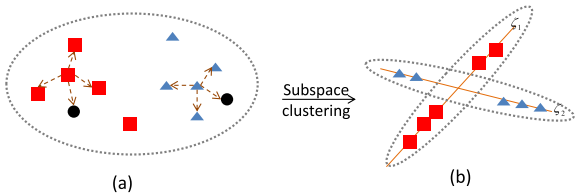


Fig. 1. Illustration of our idea. Given the spatial structure of data based on Euclidean distance in (a), two different classes of samples "□" and "Δ," "•" denotes noise in the image, and the arrows indicate potential similarity relationships between individual pixels in the subspace. Euclidean distance is limited to exploring the relationship between two points. It ignores the global information in the image. Most importantly, it does not identify the noise and also excludes similar data outside the set neighborhood. Our idea more precisely extracted multisubspace structure of the data in (b).

To mitigate the above-mentioned problems, we present a ground-breaking unmixing framework named *subspace clustering constrained sparse NMF* (SC-NMF) for hyperspectral unmixing. In this article, subspace clustering is embedded into the NMF framework, and both of them are interdependent and updated as a whole to improve the precision of unmixing. Specifically, instead of using the Euclidean distance, we utilize a self-expression coefficient matrix derived from subspace clustering to construct a similarity graph. Within the same subspace, each pixel can be expressed linearly by all other pixels. The pixel whose expression coefficient is larger indicates that the degree of similarity with the current pixel is higher. Fortunately, this similarity graph based on the reconstruction coefficients can accurately capture the multisubspace information between data points.

In addition, the idea of subspace clustering is used to generate endmembers based on the image itself by linearly combining the raw hyperspectral data. As a result, it can link the endmember spectra to the materials in HSI, ensuring the physical interpretability of the endmembers. As far as we know, this kind of unified framework is proposed for the first time. The contributions of the proposed SC-NMF can be summarized as follows.

1) A unified unmixing framework named SC-NMF for hyperspectral unmixing is proposed. In this framework, a similarity graph based on reconstruction coefficients, which can mine multisubspace structures of data, is designed to directly guide NMF for unmixing.
2) To mitigate the effects of noise in the image, a local constraint is introduced to the subspace clustering term. This constraint is that only k largest values are retained in the representation vector, which will be used to build the similarity graph.
3) The idea of subspace clustering is also applied to extract endmembers by constructing a convex hull in subspace. The process of endmember extraction can be seen as looking for archetypes of the convex hull, which can be very effective to avoid the generation of artificial endmembers.

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The rest of this article is organized as follows. Section II briefly introduces relevant background of our proposed method. In Section III, our SC-NMF method for hyperspectral unmixing will be described in detail. Section IV shows and analyses experiment results on synthetic and real HSIs. The conclusions are stated in Section V.

II. Related Works

In this section, two relevant concepts will be briefly introduced. Section II-A introduces the LMM for HSI, and Section II-B describes the nonnegative subspace clustering for capturing spatial structure of data.

A. LMM

LMM assumes that each pixel in the HSI can be seen as a linear combination of the endmembers. Given hyperspectral data $Y = [y_1, y_2, \ldots, y_N] \in \mathbb{R}^{L\times N}$, which contains L bands and N pixels. If y_n represents a pixel in the image, the LMM can be expressed as follows:

$$y_n = Ma + \varepsilon \tag{1}$$

where $M = [m_1, m_2, \ldots, m_P] \in \mathbb{R}^{L\times P}$ contains P endmembers. $a \in \mathbb{R}^{P\times 1}$ is the coding coefficient, also called abundance, and $\varepsilon \in \mathbb{R}^{L\times 1}$ represents the additive error in the image. Under this model, there are two important constraints, i.e., *abundance nonnegativity constraint* (ANC) and *abundance sum-to-one constraint* (ASC), can be given by

$$\begin{cases} a_{\text{in}} \geq 0, \quad i = 1, 2, \ldots, P \\ \sum\limits_{i=1}^{P} a_{\text{in}} = 1 \end{cases}. \tag{2}$$

Therefore, this idea is applied to the entire hyperspectral data and can be embodied in formulas as

$$Y = MA + E \tag{3}$$

where $A = [a_1, a_2, \ldots, a_n] \in \mathbb{R}^{P\times N}$ represents the abundance coefficient matrix. and $E \in \mathbb{R}^{L\times N}$ is the random additional error in the image. In summary, the objective function using the standard NMF for unmixing can be written as

$$\min_{M,A\geq 0} ||Y - MA||_F^2 \tag{4}$$

where $\|\cdot\|_F$ is the Frobenius norm.[1] Using the updated algorithm in [53], we can get

$$\begin{aligned} M &= M \odot \frac{YA^T}{MAA^T} \\ A &= A \odot \frac{M^T Y}{M^T MA} \end{aligned} \tag{5}$$

where $\odot$ represents the elementwise multiplication, and $(\cdot)^T$ is the transpose of matrix. The multiplication update rule can guarantee to find at least locally optimal solution of (4). Unfortunately, (4) has obvious nonconvex features, and many local minimums will appear in the iterative process. This phenomenon will cause instability in the result of the unmixing. Therefore, the constraints on the basic NMF have become the mainstream trend of the development.

Among the many constraints, sparse constraints on the abundance matrix are the most common operations. Especially in [42], the $L_{1/2}$ regularizer can obtain more sparse results comparing with other sparse NMF methods and the $L_{1/2}$-NMF is given as

$$\min_{M,A\geq 0} ||Y - MA||_F^2 + \lambda ||A||_{\frac{1}{2}} \tag{6}$$

where λ is a weight used to balance sparse constraints and reconstruction errors. In particular, the definition of $||A||_{1/2}$ is

$$||A||_{\frac{1}{2}} = \sum_{i=1}^{P}\sum_{j=1}^{N} A_{i,j}^{\frac{1}{2}}. \tag{7}$$

However, the solution of (6) mostly depends on the initial of M and A. When given a different initial M, A, the result of the unmixing is unstable. In addition, (6) is also very sensitive to noise. Therefore, although $L_{1/2}$-NMF has a better effect in ensuring the sparsity of the abundance matrix, there are still more constraints that need to be considered in order to improve the unmixing performance.

B. Subspace Clustering

Subspace clustering is a very useful way to explore the latent characteristic structure of the data. It aims at finding the optimal subspace of points, in which each data can be expressed as a linear combination of other points. In the past few years, there are many approaches proposed in the area of linear subspace clustering. Generally, these approaches can be roughly divided into three categories: higher order model-based methods, factorization methods, and self-expressiveness-based methods. Among all these subspaces clustering approaches, the utilization of self-expressiveness-based methods is quite popular since they are often more robust to outliers and noise than others when relying on regularization terms to account for data corruption. In essence, self-expressiveness-based methods contain both local and global models. The global model aims to exploit a similarity matrix that best describes the multilayered latent structure between all data. The local model is not only sensitive to the near neighborhood size but also has a not strong capture capability, facing the points around the intersections of different subspace since it depends on the pairwise distance-based similarity matrix [54].

As is known to all, the low-rank representation (LRR) [55] by exploring an LRR, seeking a sparse representation and sparse subspace clustering (SSC) [56], which tries to explore an LRR for data, are the two most representative global subspace clustering methods. Specifically, suppose $Y = [y_1, y_2, \ldots, y_N] \in \mathbb{R}^{L\times N}$ is the whole data set matrix, each $y_n \in \mathbb{R}^{L\times 1}$ represents a sample vector, L denotes the dimensionality of the data, and N is the total number of data. For each sample point, its linear self-expression form on the dictionary Y is

$$y_n = Ys_n, \text{ s.t. } s_{nn} = 0 \tag{8}$$

[1] In this article, unless otherwise specified, all matrices and vectors satisfy nonnegative constraints.

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where s_n is the reconstruction coefficient of y_n on the dictionary Y. The constraint $s_{nn} = 0$ is to avoid invalid solution of objective function.

By rewriting (8), we can get matrix formulation for all samples

$$Y = YS + E, \quad \text{s.t. } \text{diag}(S) = 0 \tag{9}$$

where $S = [s_1, s_2, \ldots, s_N] \in \mathbb{R}^{N\times N}$ is the self-representation matrix. E is the error matrix and diag$(\cdot)$ means the diagonal elements of the matrix. Once the subspace structure matrix S is obtained by solving (9), we can build a similarity matrix $W \in \mathbb{R}^{N\times N}$ based on $W = (|S|^T + |S|)/2$. The similarity matrix will then guide the clustering of the data.

III. Subspace Clustering-Based NMF for Hyperspectral Unmixing

In this section, we propose an integrated framework combining subspace clustering and NMF for hyperspectral unmixing to fully explore the data multisubspace information and make the endmembers interpretable. In addition, the optimization strategy of the objective function will be discussed.

A. Supergraph Learning

Based on the graph learning theory [57], we can know that a simple graph can be used to illustrate the spatial relationship between discrete data samples. In this graph model, the Euclidean distance between two points is usually used to explore their relationship, and the corresponding similarity matrix is constructed to describe the spatial information of data. Such a simple graph often loses some complex relationships between data that are important to particular applications [58]. For example, whether there are similarities between groups in the data rather than just between pairs of samples. In addition, the Euclidean distance is more affected by noise when expressing the relationship between pairs of samples. Therefore, in our article, we attempt to construct a supergraph different from the simple graph by subspace clustering method, which can fully exploit the multilayer spatial structure between hyperspectral data. Here we first introduce some basic concepts.

Given the data $Y = \{y_1, y_2, \ldots, y_N\}$, a supergraph $G = (Y, \zeta)$ describing the spatial relationship between pixels in Y is created, where $\zeta = \{\varsigma_1, \varsigma_2, \ldots, \varsigma_r\}$ denotes the subspace collection to which the current pixel y_n belongs to. In addition, G has two properties:

1) $\varsigma_j \neq \emptyset$, $for\ j = 1, 2, \ldots, r$;
2) $\varsigma_1 \cup \varsigma_2, \ldots, \cup\varsigma_r = Y$.

In G, y_n represents the vertex of the supergraph, and ς_j is called a superedge. Fig. 2 shows a simple example of the supergraph $G^* = (Y^*, \zeta^*)$, where $Y^* = \{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$, $\zeta^* = \{\varsigma_1 = \{y_1, y_2, y_3\}, \varsigma_2 = \{y_3, y_4, y_5, y_6\}, \varsigma_3 = \{y_6, y_7, y_8, y_9\}\}$. Most notably, each superedge ς can be given a positive weight $\omega(\varsigma)$, also called as the superedge weight. In our article, the weight is used to describe the similarity between data samples. Later, we will detail how to obtain this weight from the subspace clustering method.

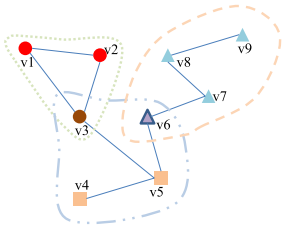


Fig. 2. Illustration of the supergraph G. The solid nodes represent the vertices of the supergraph and the dotted line region node set marked with different colors is the superedge.

B. Modeling of SC-NMF

Following the ideas in Section III-A, we propose a unified framework based on subspace clustering and NMF to jointly utilize the global spatial structure information and sparsity of HSI. In our article, the self-expression is used to describe the similarity of data in each subspace, potentially dividing the data into multiple spaces. The similarity is not directly calculated from the Euclidean distance between the two independent data but is constructed by the self-expression coefficient matrix of data. In addition, the data show their interdependence in the process of self-expression. We will explore this relationship as the subspace structure information inside the data in this article. Specifically, in our model, subspace clustering is used to construct a matrix, which describes the similarity among data. The similarity matrix is represented by self-expression coefficients since the reconstruction-based matrix can be used to extract the multisubspace structure of data. In each subspace class, each pixel can be linearly expressed by all other pixels. In this article, we can understand that the expression coefficients based on raw data should also be similar when two pixels in the image are similar. Therefore, we combine subspace clustering with NMF by a Laplace graph regularizer. Write it as a mathematical expression

$$\begin{aligned} J_L &= \frac{1}{2}\sum_i \sum_j (a_i - a_j)^2 W_{i,j} \\ &= \text{Tr}(ADA^T) - \text{Tr}(AWA^T) = \text{Tr}(ALA^T) \\ &\quad \text{s.t. } L = D - W, D_{ii} = \sum_j W_{ij} \\ &\quad i = j = 1, 2, \ldots, N \end{aligned} \tag{10}$$

where $W \in \mathbb{R}^{N\times N}$ denotes the similarity matrix between the points in the HSIs, D is a degree matrix, and Tr$(\cdot)$ represents the trace of a matrix. In this article, the subspace clustering self-expression coefficient matrix S is used to construct the similarity matrix W

$$W = \frac{|S|^T + |S|}{2}. \tag{11}$$

Therefore, the objective function of unmixing in our article can be formulated as

$$\begin{aligned} &\min_{M,A\geq 0} ||Y - MA||_F^2 + ||Y - YS||_F^2 + \lambda_1 \text{Tr}(ALA^T) + \lambda_2 \|A\|_{\frac{1}{2}} \\ &\text{s.t. } \text{diag}(S) = 0 \end{aligned} \tag{12}$$

where λ_1 and λ_2 are the parameters to balance the weight of Laplacian regularizer and sparsity item. As for the rest,

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the first term and the second term are the objection function of the standard NMF and subspace clustering, respectively. In addition, in order to better optimize our proposed objective function in the future, we add a nonnegative constraint on the self-expression coefficient matrix S in the subspace clustering. Then the framework can be rewritten as

$$\min_{M,A,S\geq 0} ||Y-MA||_F^2+||Y-YS||_F^2+\lambda_1 \mathrm{Tr}(ALA^T)+\lambda_2\|A\|_{\frac{1}{2}} \quad \text{s.t. } \mathrm{diag}(S)=0 \tag{13}$$

where the similarity matrix

$$W=\frac{S^T+S}{2}. \tag{14}$$

To improve the robustness of the proposed method, only the largest k values are remained for each column of S before constructing the similarity matrix W. It can be implemented using the threshold operation in [59].

C. Subspace Clustering for Endmembers Extraction

Considering the interpretability of NMF results, the endmember spectra will be difficult to relate to the real objects in the image. Specifically, when unmixing is viewed as a blind source separation problem and mathematical formulas are used to solve it, the unreliability of the solution will inevitably arise. In this article, we propose to use the idea of subspace clustering to simulate the endmembers by a linear combination of all the raw pixels inside the subspace, which can be used to avoid the generation of artificial endmembers and ensures that the final results are as consistent as possible with real scenes. Expressed in mathematical form as

$$m_j=\sum_{i=1}^{N} b_{i,j}y_i \quad \text{s.t. } b_{i,j}\geq 0,\quad i=1,2,\ldots,N,\ \ j=1,2,\ldots,D \tag{15}$$

where y_i is a spectral vector of the ith pixel in the spectral space and $b_{i,j}$ denotes the weight coefficient of ith pixel when m_j endmember is linearly expressed. $b_{i,j}$ can also be interpreted as the possibility that ith pixel represents the endmember m_j. In addition, the meanings of the nonnegative constraint and the sum-to-one constraint are the same as the constraints of abundance. This idea in matrix form can be rewritten as

$$M=YB \quad \text{s.t. } B\geq 0 \tag{16}$$

where B is a coefficient matrix that represents the endmembers with raw data. Therefore, the objective function (13) can be rewritten as

$$\min ||Y-YBA||_F^2+||Y-YS||_F^2+\lambda_1 \mathrm{Tr}(ALA^T)+\lambda_2\|A\|_{\frac{1}{2}} \quad \text{s.t. } B,A,S\geq 0,\quad ||a_j||_1=1,\ \mathrm{diag}(S)=0. \tag{17}$$

D. Optimization

In this section, we will describe the process of optimization of the objective function (17) in detail, which is evidently not convex when talking about three matrices, i.e. B, A, and S, together. Therefore, we use an iterative method to solve them.

When S is fixed, the objection function (17) can be represented

$$\min_{B,A\geq 0} \|Y-YBA\|_F^2+\lambda_1 \mathrm{Tr}(ALA^T)+\lambda_2\|A\|_{\frac{1}{2}} \quad \text{s.t. } B,A\geq 0,\quad ||a_j||_1=1. \tag{18}$$

We can see that only two parameters in the above expression are variables. Therefore, as suggested in [58], we deduce the update rules of B and A by the method of Lagrange multipliers.

To facilitate the solution and display, we rewrite the formula (18) as

$$\begin{aligned} f(B,A) &= \mathrm{Tr}((Y-YBA)(Y-YBA)^T)+\lambda_1\mathrm{Tr}(ALA^T)+\lambda_2\|A\|_{\frac{1}{2}} \\ &= \mathrm{Tr}(YY^T)-2\mathrm{Tr}(YBAY^T)+\mathrm{Tr}(YBAA^TB^TY^T)+\lambda_1\mathrm{Tr}(ALA^T)+\lambda_2\|A\|_{\frac{1}{2}}. \end{aligned} \tag{19}$$

According to the known constraints, we can get: $B^{N\times P}\geq 0$ and $A^{P\times N}\geq 0$. Let $\Psi^{N\times P}\geq 0$ and $\Phi^{P\times N}\geq 0$ both represent the Lagrangian multipliers, then the following formulas Γ :

$$\Gamma=\mathrm{Tr}(YY^T)-2\mathrm{Tr}(YBAY^T)+\mathrm{Tr}(YBAA^TB^TY^T)+\lambda_1\mathrm{Tr}(ALA^T)+\lambda_2\|A\|_{\frac{1}{2}}+\mathrm{Tr}(\Psi A^T)+\mathrm{Tr}(\Phi B^T). \tag{20}$$

Find the partial derivatives of A and B for the above formula, and obtain

$$\begin{aligned} \frac{\partial\Gamma}{\partial B} &= -2Y^TYA^T+2Y^TYBAA^T+\Psi \\ \frac{\partial\Gamma}{\partial A} &= -2B^TY^TY+2B^TY^TYBA+2\lambda_1AL+\frac{1}{2}\lambda_2A^{-\frac{1}{2}}+\Phi. \end{aligned} \tag{21}$$

In [60], we have learned about *Karush–Kuhn–Tucker* (KKT) conditions

$$\begin{cases} \Psi_{np}B_{np}=0 \\ \Phi_{pn}A_{pn}=0. \end{cases} \tag{22}$$

Applying it to (21), we can obtain

$$(-2Y^TYA^T+2Y^TYBAA^T)_{nr}\odot B_{np}=0 \tag{23}$$

$$\left(-2B^TY^TY+2B^TY^TYBA+2\lambda_1AL+\frac{1}{2}\lambda_2A^{-\frac{1}{2}}\right)_{pn}\odot A_{pn}=0. \tag{24}$$

From the Laplacian theory, we know: $L=D-W$, together with applying the transposition and division method, the following updated rules for unknown variables can be acquired:

$$B=B\odot\frac{Y^TYA^T}{Y^TYBAA^T} \tag{25}$$

$$A=A\odot\frac{4B^TY^TY+4\lambda_1AW}{4B^TY^TYBA+\lambda_1AD+\lambda_2A^{-\frac{1}{2}}}. \tag{26}$$

At the same time, when both B and A are fixed, the objective function (17) can be reduced as follows:

$$\min_{S\geq 0} ||Y-YS||_F^2+\lambda_1\mathrm{Tr}(ALA^T). \tag{27}$$

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According to [61], the above formula can be rewritten as

$$\min_{S\geq 0} ||Y - YS||_F^2 + \lambda_1 \mathrm{Tr}(S^T P) \tag{28}$$

where $P_{i,j} = ||a_i - a_j||_2^2$, a_i is the ith column of coefficient matrix A. Finally, the rule of updating S is the same as B and A. By resorting to the KKT conditions, the transposition and division, the updating rule for S can be obtained as

$$S = S \odot \frac{4Y^T Y}{4Y^T YS + \lambda_1 P}. \tag{29}$$

Then, we can calculate the similarity matrix W based on S: $W = (S^T + S)/(2)$ and determine the Laplacian map L based on the similarity matrix: $L = D - W$. It is noteworthy that we use threshold operations to keep the largest k for each column of S when we calculate the self-expressing matrix S. The advantage is that the mutual interference between the cluster subspaces and noise can be reduced as much as possible.

The convergence of the update rules (25), (26), and (29) is proved by Theorem 1.

Theorem 1: Under the update rules (25), (26), and (29), the objective function in (17) is nonincreasing.

A detailed proof of Theorem 1 is shown in the Appendix, and its proof is similar to the proof of convergence in [53].

E. Implementation Issues

In this section, the implementation details of the proposed algorithm are discussed. The first issue is how to guarantee ANC and ASC constraints in LMM. In this article, it is obvious that the SC-NMF algorithm satisfies the ANC constraint as long as the initial matrix is nonnegative. For the ASC constraint, we use the method proposed in [42] to ensure it. Specifically, when updating the abundance A, the HSI data Y and the endmember $M = YB$ are simultaneously augmented as

$$Y_n = \begin{bmatrix} Y \\ \delta 1_N^T \end{bmatrix}$$
$$M_n = \begin{bmatrix} M \\ \delta 1_D^T \end{bmatrix} \tag{30}$$

where δ is used to control the impact of the ASC. According to the recommendation in [62], δ is set to 15 in the experiments.

Another issue is how the algorithm stops updating. In general, there are two methods to choose from. One is to set an error tolerance, and once the loss error is more than ten times within the tolerance, the iteration is terminated [34]. Another way is to set a maximum number of iterations. In this article, we set the number of algorithm iterations to 3000 times, which can fully guarantee the algorithm convergence.

The SC-NMF algorithm proposed in this article can be summarized in Algorithm 1.

Algorithm 1 SC-NMF for Hyperspectral Unmixing

Input:
The hyperspectral data $\mathbf{Y} \in \mathbb{R}^{L\times N}$, the number of endmembers P, the number of $k > 0$, and the parameters $\lambda_1 > 0$, $\lambda_2 > 0$.

Output:
Endmembers matrix **M**, abundance matrix **A**.

Initialize B and **A** with nonnegative random matrices, **S** and **L** are initialized by applying Heat kernel matrix.

repeat

1:Update **B**, **A**, **S** by applying the updating rules in equation (25),(26) and (29);

2:Keep only k largest values in each column of **S**;

3:Update similarity matrix W and laplacian matrix L;

until Converges.

4:Calculate endmembers matrix **M**: $\mathbf{M} = \mathbf{YB}$.

IV. Experiments

In this section, several experiments on both synthetic and real hyperspectral data are designed to evaluate the effectiveness of the proposed SC-NMF algorithm. In order to fully demonstrate the performance of our method, here we compare it with the five classic algorithms, which are CSNMF, robust constrained matrix factorization (RCMF) [19], $L_{1/2}$-NMF, simplex identification via split augmented Lagrangian (SISAL-FCLS) [63], and VCA-FCLS. In these algorithms, CSNMF considers the spatial information in unmixing.

For all experiments, we use the two most commonly used matrices, including spectral angle distance (SAD) and root-mean-square error (RMSE), as evaluation parameters to illustrate the performance of each comparison algorithm. SAD can be used to calculate the angle distance between the endmember matrix solved by the algorithm M_p and the real endmember matrix $\hat{M}_p$, which p is pth endmember. The specific calculation method can be mathematically expressed as

$$\mathrm{SAD}_p = \cos^{-1}\left(\frac{\hat{M}_p^T M_p}{\|\hat{M}_p\|\|M_p\|}\right). \tag{31}$$

RMSE denotes the error of the abundance matrix solved by the algorithm A_p and the ground-truth abundance matrix $\hat{A}_p$. Specifically

$$\mathrm{RMSE}_p = \left(\frac{1}{n}|A_p - \hat{A}_p|^2\right)^{\frac{1}{2}}. \tag{32}$$

All experiments are repeated 20 times under the same conditions, and the average is used as the final experimental result.

A. Experiments on Synthetic Data

The endmember spectra required in the experiments are all from the USGS spectral library. Fig. 3 shows the spectra of six materials. In order to generate synthetic data similar to the real scene image, we use the method described in [50]. The specific steps are given as follows. 1) Divide the image into small blocks of $d \times d$, when the size of the image to be generated is $d^2 \times d^2$. 2) Randomly fill the pure endmember into each small piece and ensure that each piece contains only one substance. 3) Degrade the generated image with a low-pass filter of $(d-1)\times(d-1)$. 4) Replace pixels whose abundance is greater than the predefined threshold θ with

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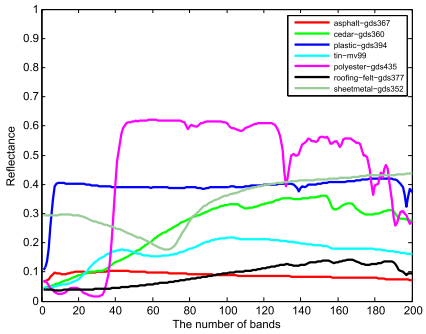


Fig. 3. Example endmember spectral plots in synthetic data.

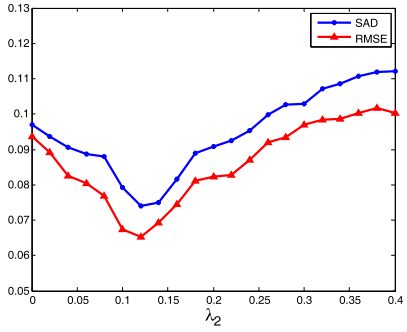


Fig. 5. Performance of SC-NMF with respect to parameter λ_2.

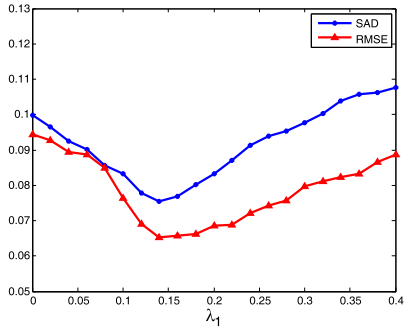


Fig. 4. Performance of SC-NMF with respect to parameter λ_1.

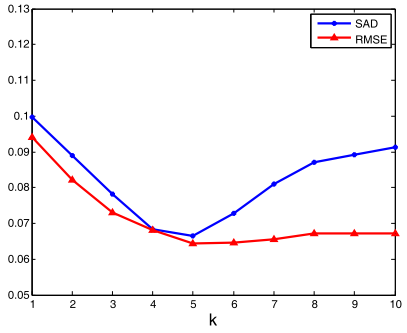


Fig. 6. Performance of SC-NMF with respect to parameter k.

pixels consisting of equal abundances of all endmembers to guarantee that all pixels are highly mixed. Among them, θ can be used to control the degree of mixing of pixels, where the larger the theta, the lower the degree of pixel mixing. 5) Add zero-mean white Gaussian noise to the synthetic data to simulate HSIs as realistically as possible.

1) Experiment 1 (Parameter Setting): There are three parameters in our proposed method, including λ_1, λ_2, and k. In this experiment, we will analyze these parameters to optimize the performance of our algorithm. When synthesizing data, we set the number of endmembers $P = 4$, image size is 64×64, and the degree of endmembers $\theta = 0.6$, and $SNR = 20$. All experiments allow the control variable method to determine a parameter when fix others and use SAD and RMSE as indicators to measure the effect of unmixing. The values of the parameters are set as: λ_1 ={0, 0.05, 0.1, 0.15, 0.2 0.25, 0.3, 0.35, 0.4}, λ_2 ={0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4}, and k ={1, 2, 3, 4, 5, 6, 7, 8, 9, 10}.

In all test experiments, in order to avoid the impact of random initialization of the matrix on results, we made all matrix initializations be the same. First, keep $\lambda_2 = 0.1$ based on the experience in [34] and fix $k = 4$. The unmixing results obtained are shown in Fig. 4. As we can see from it, SAD and RMSE tend to be consistent with the change in λ_1. When λ_1 is less than 0.14, the performance of the proposed algorithm is continuously enhanced with the increase in λ_1, and when λ_1 is greater than 0.14, both SAD and RMSE remain positively correlated with λ_1, indicating that the unmixing effect is deteriorated.

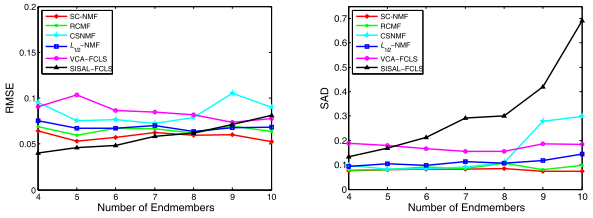
Second, by fixing $\lambda_1 = 0.14$ and $k = 4$, the influence of the value of λ_2 on the performance of the proposed algorithm is explored. The results are presented in Fig. 5. We can conclude that both indicators take the minimum value when $\lambda_2 = 0.12$. When λ_2 is greater than or less than 0.12, the corresponding SAD and RMSE increase largely. In other words, $\lambda_2 = 0.12$ is the best coefficient of abundance sparse constraint.

Third, $\lambda_1 = 0.14$ and $\lambda_2 = 0.12$ are fixed and select the optimal value k. Fig. 6 shows the results of two matrices varying with k. During the process of k changing from 1 to 5, both SAD and RMSE are continuously decreasing, and when the variable range of k is 5 to 10, the RMSE tends to be stable and the SAD is constantly increasing. Therefore, $k = 5$ is the parameter value that makes SC-NMF optimal. In summary, we set $\lambda_1 = 0.14$, $\lambda_2 = 0.12$ and $k = 5$ in all subsequent experiments since only then can our method SC-NMF achieves the highest accuracy of endmembers and abundances after unmixing.

2) Experiment 2 (Robustness to Different Number of Endmembers): This experiment is used to explore the effect of the difference in the number of endmembers on the unmixing performance of the proposed method and the other five methods. When generating synthetic data, we set $\theta = 0.6$, the number of endmembers P to vary from 4 to 10. It is worth noting that in the process of changing the number of endmembers, all comparison algorithms are guaranteed to use the same synthetic data at the same time. The test results are shown in Fig. 7. Comprehensive comparison results, we can confirm that our algorithm has strong superiority and stability.

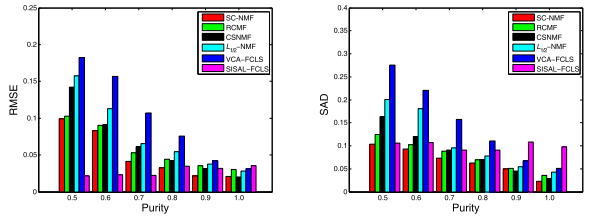
3) Experiment 3 (Generalization to Different Mixing Degree): We know that when the pixels in the image have

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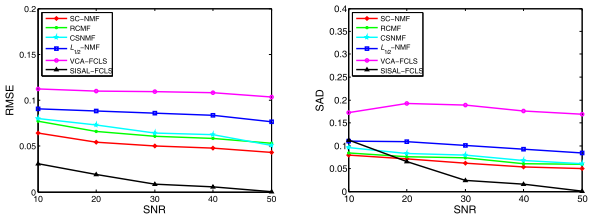
(a) (b)

Fig. 7. Comparison of the algorithms with different numbers of endmembers in terms of (a) RMSE and (b) SAD.



(a) (b)

Fig. 8. Comparison of the algorithms with different mixing degrees in terms of (a) RMSE and (b) SAD.

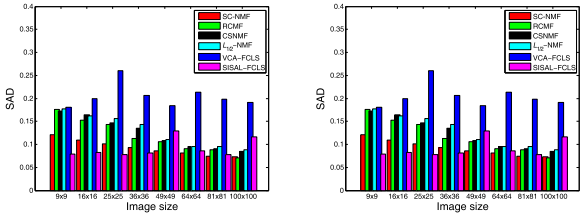


(a) (b)

Fig. 9. Comparison of the algorithms at different noise levels in terms of (a) RMSE and (b) SAD.

different degrees of mixing, the unmixing performance of algorithms is discrepant. The actual object itself is intricate, so it is very necessary to discuss the robustness of the algorithm to the endmember mixture. To evaluate the results, when the data are synthesized, the other parameters are fixed, and the metric θ of the degree of mixing is adjusted to increase from 0.5 to 1 at intervals of 0.1. According to the aforementioned, we know that the larger θ is, the lower the degree of mixing of pixels. When $\theta = 1$, in the ideal case, the pixels are pure. For each type of synthetic data, there are five different algorithms that are compared to our algorithm. As shown in Fig. 8, with the pixel mixing degree increasing, the performance of all algorithms decreases, but the proposed SC-NMF algorithm has the best performance in both SAD and RMSE.

4) Experiment 4 (Generalization of Noise): In the acquisition of HSIs, noise is always accompanied. Therefore, a suitable unmixing method must accommodate the presence of different errors. In this experiment, we set $\theta = 0.6$, the value of $SNR = 10, 20, 30, 40, 50$, respectively, which means that noise in the image is sequentially reduced. The results of various comparison algorithms are shown in Fig. 9. They demonstrate that the performance of each



(a) (b)

Fig. 10. Comparison of the algorithms with different image sizes in terms of (a) RMSE and (b) SAD.

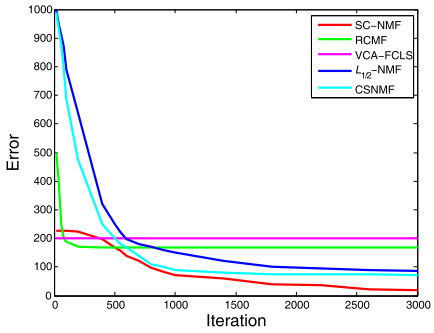


Fig. 11. Comparison of the algorithms in terms of an approximation error.

algorithm is gradually increased when the noise in the image is sequentially reduced. Among them, $L_{1/2}$-NMF is the most stable, and its performance hardly changes with the vary of SNR. When the SNR is greater than 20, the performance of the SISAL is the best while the VCA is the worst. Compared with RCMF and CSNMF, our algorithm can highlight certain advantages. A comprehensive analysis of the results shows that SC-NMF has both stability and good results to a certain extent.

5) Experiment 5 (Different Image Size Analysis): In this experiment, we evaluate the generalization capabilities of the six methods for image unmixing at different sizes. In the process of synthesizing the data, we set $\theta = 0.6$, and let $d = 3, 4, 5, 6, 7, 8, 9, 10$ be made into eight images of sizes 9×9, 16×16, 25×25, 36×36, 49×49, 64×64, 81×81, 100×100, respectively. Fig. 10 shows that the performance of SC-NMF is getting better as the image size increases. It is mainly because after subspace clustering, more spatial structure information is explored to assist the unmixing. CSNMF, RCMF, and $L_{1/2}$-NMF all have fairly good effects as the image size continues to increase, whereas VCA and SISAL show an unstable trend. Therefore, with comprehensive stability and unmixing accuracy, SC-NMF has slightly superior performance.

6) Experiment 6 (Convergence Analysis): In practical applications, the timeliness of the algorithm needs to be considered. Therefore, here we discuss the convergence speed of four methods, including $L_{1/2}$-NMF, VCA-FCLS, RCMF, and SC-NMF. The variation in the approximation error of the four algorithms with the increase in the number of iterations is shown in Fig. 11. We can learn that the VCA-FCLS error during the whole iteration is constant since VCA-FCLS is deterministic. The convergence speed of RCMF is the

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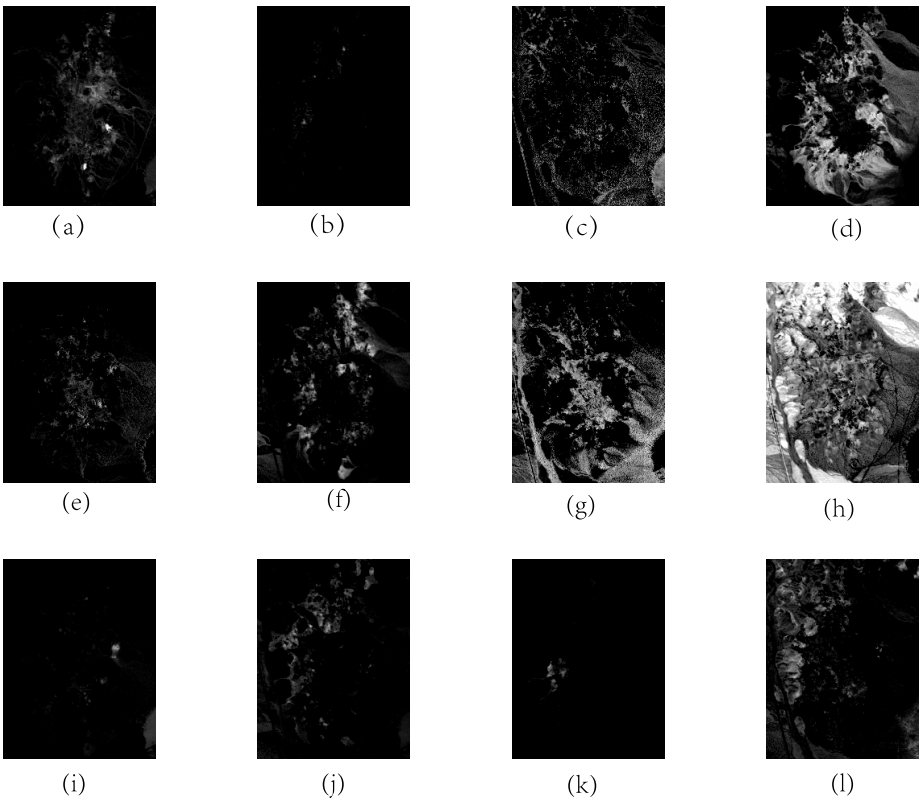


Fig. 12. Abundance maps of different endmembers using SC-NMF on the AVIRIS Cuprite data. (a) Dumortierite. (b) Kaolinite #2. (c) Nontronite. (d) Alunite. (e)Chalcedony. (f) Kaolinite #1. (g)Andradite. (h) Sphene. (i) Muscovite. (j) Montmorillonite. (k) Buddingtonite. (l) Pyrope.

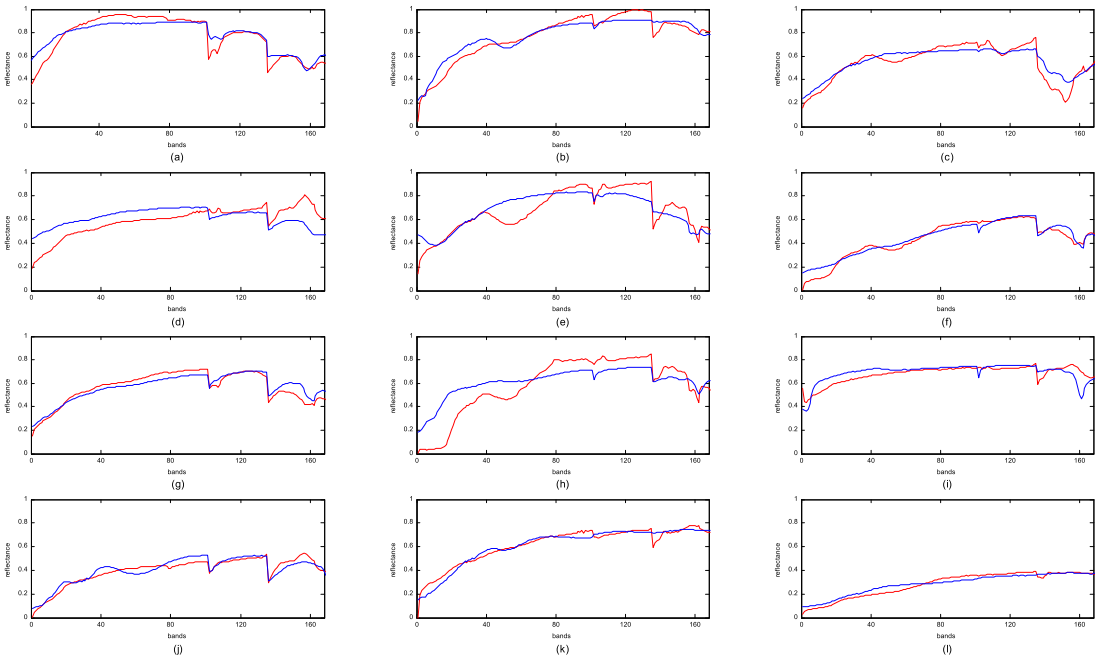


Fig. 13. Comparison of the USGS library spectra (red line) with the endmember signatures extracted by SC-NMF (blue line) on the AVIRIS Cuprite data. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Chalcedony. (e) Dumortierite. (f) Kaolinite #1. (g) Kaolinite #2. (h) Montmorillonite. (i) Muscovite. (j) Nontronite. (k) Pyrope. (l) Sphene.

fastest, and it has converged after about 100 iterations. Our method demonstrates that the convergence of the algorithm is slightly slower, but its approximation error can reach the minimum value as the number of iterations increases, thus making the endmember and abundance values the most accurate.

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TABLE I
SAD AND THE STANDARD VARIANCE RESULTS ON THE AVIRIS CUPRITE DATA

	RCMF	CSNMF	$L_{1/2}$-NMF	VCA-FCLS	SISAL-FCLS	SC-NMF
Alunite	0.1027±0.0248	0.1054±0.0216	0.1039±0.0682	0.1129±0.0814	0.2054±0.7641	**0.0982±0.0313**
Andradite	0.0835±0.0670	0.0844±0.0432	**0.0822±0.0131**	0.0886±0.0276	0.0992±0.5604	0.0831±0.0170
Buddingtonite	0.1054±0.0119	0.1123±0.0172	0.0998±0.0134	0.0900±0.0181	0.1217±0.2581	**0.0867±0.0319**
Chalcedony	0.0916±0.0363	**0.0832±0.0541**	0.1307±0.0734	0.1296±0.0247	0.1316±0.0835	0.1029±0.0146
Dumortierite	0.0943±0.0725	0.1390±0.0450	0.0932±0.0416	**0.0861±0.0707**	0.1247±0.2159	0.1024±0.0218
Kaolinite#1	0.0982±0.1242	0.1147±0.0916	0.0778±0.0843	0.0790±0.0847	0.1441±0.2236	**0.0736±0.0152**
Kaolinite#2	0.0814±0.0211	0.1041±0.1841	0.0692±0.0728	0.0704±0.0524	0.0609±0.0718	**0.0548±0.0113**
Montmorillonite	0.0872±0.0845	0.0706±0.0755	0.0695±0.0531	0.0637±0.0316	**0.0636±0.0367**	0.1147±0.1053
Muscovite	0.0951±0.0690	**0.0534±0.0085**	0.1644±0.0815	0.1443±0.0924	0.1336±0.0882	0.11028±0.0733
Nontronite	0.0978±0.0188	0.1363±0.0407	0.0894±0.0141	**0.0861±0.0065**	0.1023±0.0882	0.0896±0.0065
Pyrope	0.0914±0.0388	0.0806±0.0245	0.0616±0.0126	0.0861±0.0306	0.1580±0.6267	0.0635±0.0129
Sphene	0.0893±0.0995	0.0601±0.0590	0.1628±0.1047	0.1368±0.0842	0.2128±0.2707	**0.0823±0.0231**
Mean	0.0932±0.0654	0.0953±0.0715	0.1004±0.0614	0.0978±0.0583	0.1062±0.3601	**0.0887± 0.0414**

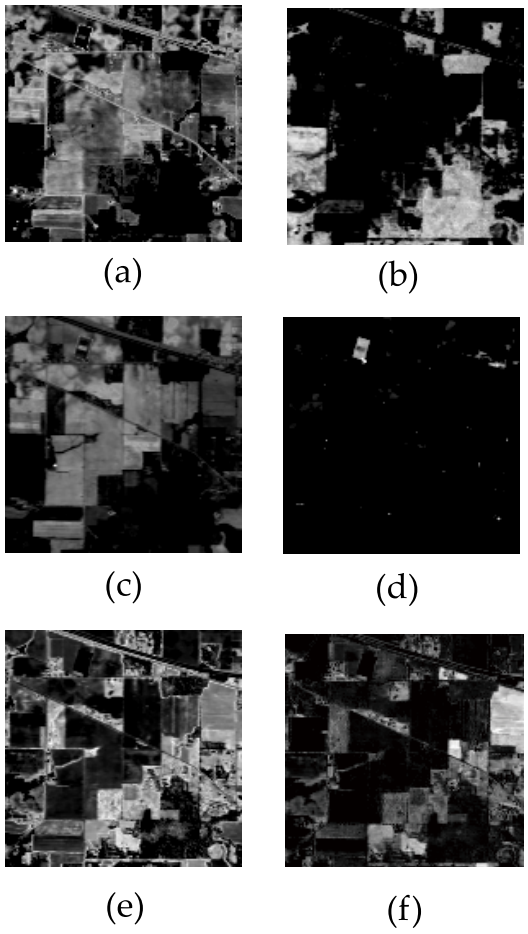


Fig. 14. Abundance maps of different endmembers using SC-NMF on the AVIRIS Indiana data. (a) Man-made land. (b) Wheat. (c) Corn. (d) Soybean. (e) Vegetation. (f) Haystack.

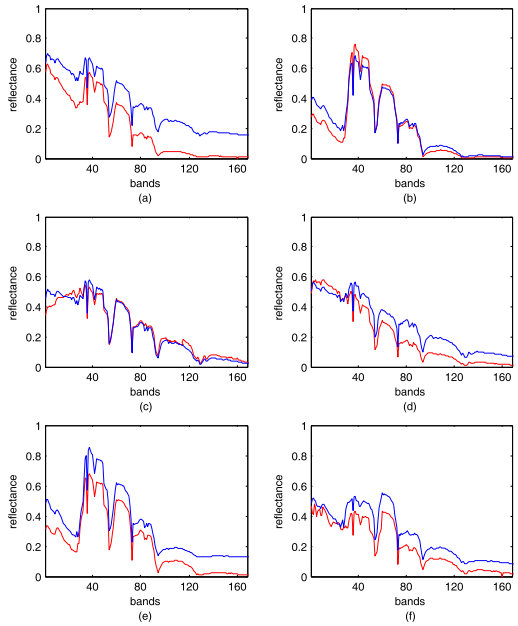


Fig. 15. Abundance maps of different endmembers using SC-NMF on the AVIRIS Indiana data. (a) Man-made land. (b) Wheat. (c) Corn. (d) Soybean. (e) Vegetation. (f) Haystack.

B. Experiments on Real Hyperspectral Data

To verify the validity of the proposed SC-NMF, we will conduct two real hyperspectral data in this section, which are captured by the *Airborne Visible/Infrared Imaging Spectrometer* (AVIRIS) sensor. The first one, a mining area with a small amount of vegetation, is called Cuprite [16]. It is widely used in hyperspectral unmixing to test the performance of various methods [16], [25], [34], [50]. Cuprite contains 224 bands with a spectral range of 0.4–2.5 μm. To improve the reliability of the experimental results, we denoise the raw data and remove the noisy band, retaining 188 bands before unmixing.

According to [16], we know that there are 14 endmembers in the Cuprite, which are Alunite, Kaolinite #1, Kaolinite #2, Sphene, Andradite, Kaolinite #3, Nontronite, Muscovite, Pyrope, Chalcedony, Kaolinite #4, Dumortierite, Montmorillonite, and Buddingtonite. Since the spectra of the four Kaolinites are very similar, we reduce the number of endmembers to 12 in the experiment. Fig. 12 shows a map of abundances on a gray-scale produced by SC-NMF. The brighter pixels in the map represent the larger abundances of the corresponding endmember in the image. As shown in Fig. 13, it is the endmembers signatures of the cuprite extracted by SC-NMF and the corresponding USGS library spectral comparison results. By analyzing the figure, we can see that the two curves are very close in most cases, that is, the extracted endmembers are basically close to the real material. Table I gives the SAD values used by six methods for the unmixing of Cuprite data, including SC-NMF, RCMF, CSNMF, $L_{1/2}$-NMF,

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TABLE II
SAD AND THE STANDARD VARIANCE RESULTS ON THE AVIRIS INDIANA DATA

	RCMF	CSNMF	$L_{1/2}$-NMF	VCA-FCLS	SISAL-FCLS	SC-NMF
Man-made land	0.1123±0.1216	0.2517±0.3817	0.1138±0.1861	**0.0924±0.0610**	0.1820±0.2634	0.3217±0.2719
Wheat	0.0618±0.1444	0.1567±0.0847	0.3412±0.0029	0.3546±0.0142	0.4466±0.2069	**0.0358±0.0096**
Corn	0.0943±0.0817	0.1068±0.0984	0.1516±0.0768	0.0761±0.1113	0.3262±0.1542	**0.0368±0.0073**
Soybean	0.0732±0.0209	0.0721±0.0130	**0.0444±0.0836**	0.1111±0.1608	0.5995±0.1450	0.0634±0.0089
Vegetation	0.1427±0.3294	0.0508±0.1498	**0.0421±0.2707**	0.0464±0.1103	0.3198±0.1733	0.1490±0.1219
Haystack	0.2308±0.4480	0.0642±0.1381	0.1658±0.0394	0.1945±0.1633	0.2059±0.1356	**0.0345±0.0236**
Mean	0.1192±0.1878	0.1052±0.2145	0.1431±0.1428	0.1458±0.1162	0.3467±0.1850	**0.0902± 0.1161**

VCA-FCLS, and SISAL-FCLS. The lowest endmembers variances are shown in bold print, from which we can conclude that the performance of SC-NMF is much better than the other five methods. Moreover, the extracted endmembers are mostly accurate in most cases, and the proposed has the smallest SAD mean.

The second is the Indiana data, which is also used [16], [25], [34], [50]. Indiana data contain 224 bands. It performs the same operation as the Cuprite data, removing the low SNR and the larger water-vapor absorption bands. Finally, only 169 bands remain for testing. As can be seen in [34], [64], there are six types of pure substances in Indiana data, including wheat, man-made land, soybean, corn, vegetation, and haystack. Since these data do not have a corresponding ground truth value in the USGS library, we use the method described in [65] to approximate the ground truth. Find a certain number of pixels close to the pure material of these six endmembers, and use their average value as the ground truth of each corresponding endmember.

As shown in Fig. 14, there are abundance maps obtained by SC-NMF unmixing. We can conclude that the abundance is consistent with the distribution of real objects in [65]. The result of comparing the solved the endmember with the ground truth is illustrated in Fig 15. By analyzing the curves in the figure, we find that our result is reliable. Table II gives the SAD for the six comparative methods. Summarizing this table, although SC-NMF does not always extract endmembers accurately, it is optimal overall and has the lowest average SAD.

V. CONCLUSION

We propose a novel approach named SC-NMF based on the joint framework of subspace clustering and sparse NMF for hyperspectral unmixing. In this framework, the spectral subspace clustering method is taken advantage of searching for the optimal subspace of each pixel to fully exploit the multisubspace structure information in the HSI. Meanwhile, the idea of subspace clustering is used to extract the endmembers by a linear combination of pixels among them. Furthermore, to improve the robustness of our method, only the largest k values are kept in clustering self-expression matrix S during each iterative optimization process, which makes our results more credible and available. We validated the proposed method both on the synthetic data and real hyperspectral data. As a result, it shows that the proposed SC-NMF can have a more remarkable performance by incorporating subspace clustering into NMF than the state-of-the-art methods.

APPENDIX
PROOF OF THE THEOREM 1

Since only the first term of the objective function is related to B, it can be easily verified that under the update rule (25), (17) is nonincreased. In addition, when B and S are fixed, the form of (17) is the same as the objective function in [58]. According to the proof in [58], the value of (17) is nonincreasing under update rule (26).

Therefore, the following work is mainly to verify that when A and B are fixed, (29) can guarantee that the objective function of problem (17) is nonincreasing. Drawing on the work in [66], we will prove the convergence of SC-NMF by introducing auxiliary functions.

First, given the definition of the auxiliary function.

Definition 1: For $F(s)$, if $J(s, s')$ satisfies the condition

$$J(s, s') \geq F(s), J(s, s) = F(s) \tag{33}$$

then $J(s, s')$ is called the auxiliary function of $F(s)$.

According to Definition 1, the following lemma can be obtained.

Lemma 1: If J is an auxiliary function of F, the value of F is nonincreasing according to the following update rule:

$$s^{t+1} = \arg\min_{s} J(s, s'). \tag{34}$$

According to Lemma 1, we will construct an appropriate auxiliary function J and prove that the update rule (29) can be obtained from (33). We rewrite (17) as follows:

$$F = \mathrm{Tr}(Y^T Y) - 2\mathrm{Tr}(SY^T Y) + \mathrm{Tr}(Y^T YSS^T) + \frac{\lambda_1}{2}\mathrm{Tr}(SP) \tag{35}$$

where P is a symmetric matrix. For each element s_{ij} in S, let F_{ij} denote the component of F that contains only s_{ij}. It is easy to obtain

$$F'_{ij} = \left(\frac{\partial F}{\partial S}\right)_{ij} = \left(-2Y^T Y + 2Y^T YS + \frac{\lambda_1}{2}P\right)_{ij} \tag{36}$$

$$F''_{ij} = 2(Y^T Y)_{ii}. \tag{37}$$

Considering that the algorithm is updated for each element, it is only necessary to prove that each F_{ij} is nonincreasing under the update rule (29).

Lemma 2: The function

$$J(s, s_{ij}^t) = F_{ij}(s_{ij}^t) + F'_{ij}(s_{ij}^t)(s - s_{ij}^t) + \frac{(Y^T YS)_{ij} + \frac{1}{4}\lambda_1 (P)_{ij}}{s_{ij}^t}(s - s_{ij}^t)^2 \tag{38}$$

is an auxiliary function of F_{ij}.

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Proof: It is easy to verify that $J(s,s) = F_{ij}(s)$. Next, we only need to prove $J(s, s_{ij}^t) \geq F_{ij}(s)$. First, the Taylor series expansion of $F_i j(s)$, we can obtain

$$F_{ij}(s) = F_{ij}\left(s_{ij}^t\right) + F'_{ij}\left(s_{ij}^t\right)\left(s - s_{ij}^t\right) + (Y^T Y)_{ii}\left(s - s_{ij}^t\right)^2. \tag{39}$$

Comparing (37) and (38), it can be seen that to prove $J(s, s_{ij}^t) \geq F_{ij}(s)$, we only need to prove

$$\frac{(Y^T Y S)_{ij} + \frac{1}{4}\lambda_1(P)_{ij}}{s_{ij}^t} \geq (Y^T Y)_{ii}. \tag{40}$$

Since

$$(Y^T Y S)_{ij} = \sum_{q=1}^{n} (Y^T Y)_{iq} s_{qj}^t \geq (Y^T Y)_{ii} s_{ij}^t \tag{41}$$

and

$$\frac{\lambda_1(P)_{ij}}{4 s_{ij}^t} \geq 0 \tag{42}$$

(39) is obviously established. Therefore, the function (37) is an auxiliary function of F_{ij}.

By asking the auxiliary function $J(s, s_{ij}^t)$ for the derivative of s and making the derivative 0, we can get that

$$F'_{ij}\left(s_{ij}^t\right) + \frac{(Y^T Y S)_{ij} + \frac{1}{4}\lambda_1(P)_{ij}}{s_{ij}^t} * 2\left(s - s_{ij}^t\right) = 0. \tag{43}$$

It is equivalent to

$$\begin{aligned} s_{ij}^{t+1} &= s_{ij}^t - s_{ij}^t \frac{F'_{ij}\left(s_{ij}^t\right)}{2(Y^T Y S)_{ij} + \frac{\lambda_1}{2} P_{ij}} \\ &= s_{ij}^t \frac{4(Y^T Y)_{ij}}{(4 Y^T Y S + \lambda_1 P)_{ij}}. \end{aligned} \tag{44}$$

Therefore, the update rule (29) is the optimal solution of the auxiliary function $J(s, s_{ij}^t)$, and it can be guaranteed that F_{ij} is nonincreasing. Theorem 1 is proved.

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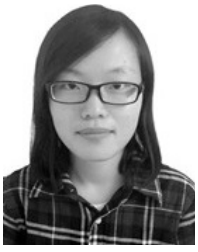
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